



RoboReviews

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46,538 reviews

→ 4 clean meta-categories

→ 4 ready-to-publish recommendation articles

Introduction

Three connected models, one coherent application

Three connected NLP models, one pipeline

Each model's output feeds the next: sentiment → clustering → article generation



Same base model (Qwen 2.5-1.5B) for Tasks 1 and 3 — efficient, consistent

Connected

Each model's output feeds the next

Reusable

Same Qwen backbone for Task 1 + 3

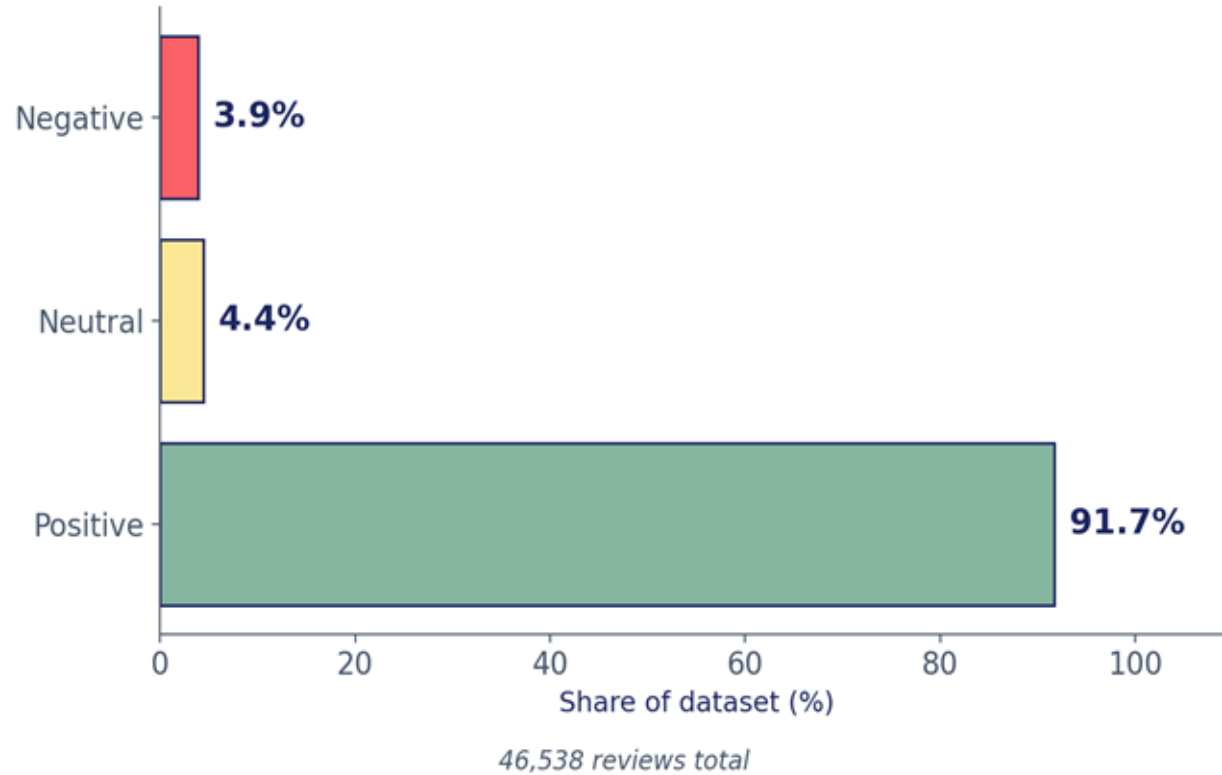
Honest

Statistical tests, held-out reference

Methods

Dataset · preprocessing · evaluation strategy

A heavily imbalanced dataset



3

datasets merged

46,538

rows after delete duplicated

**37,230 /
9,308**

split train/test

Preprocessing pipeline

Merge

Delete duplicated

Clean text

Rating → label

split

Model 1 — Sentiment Classification

Classify reviews as positive, neutral, or negative

The challenge

- **92% positive class — heavy imbalance**
- Rating-based labels are noisy: low-rating customers often write 'love it'
- Neutral class is hard — overlaps with both ends of the spectrum
- Goal: produce a usable label for downstream Tasks 2 & 3

1

Zero-Shot

Plain prompt

2

N-Shot

8 in-context examples

3

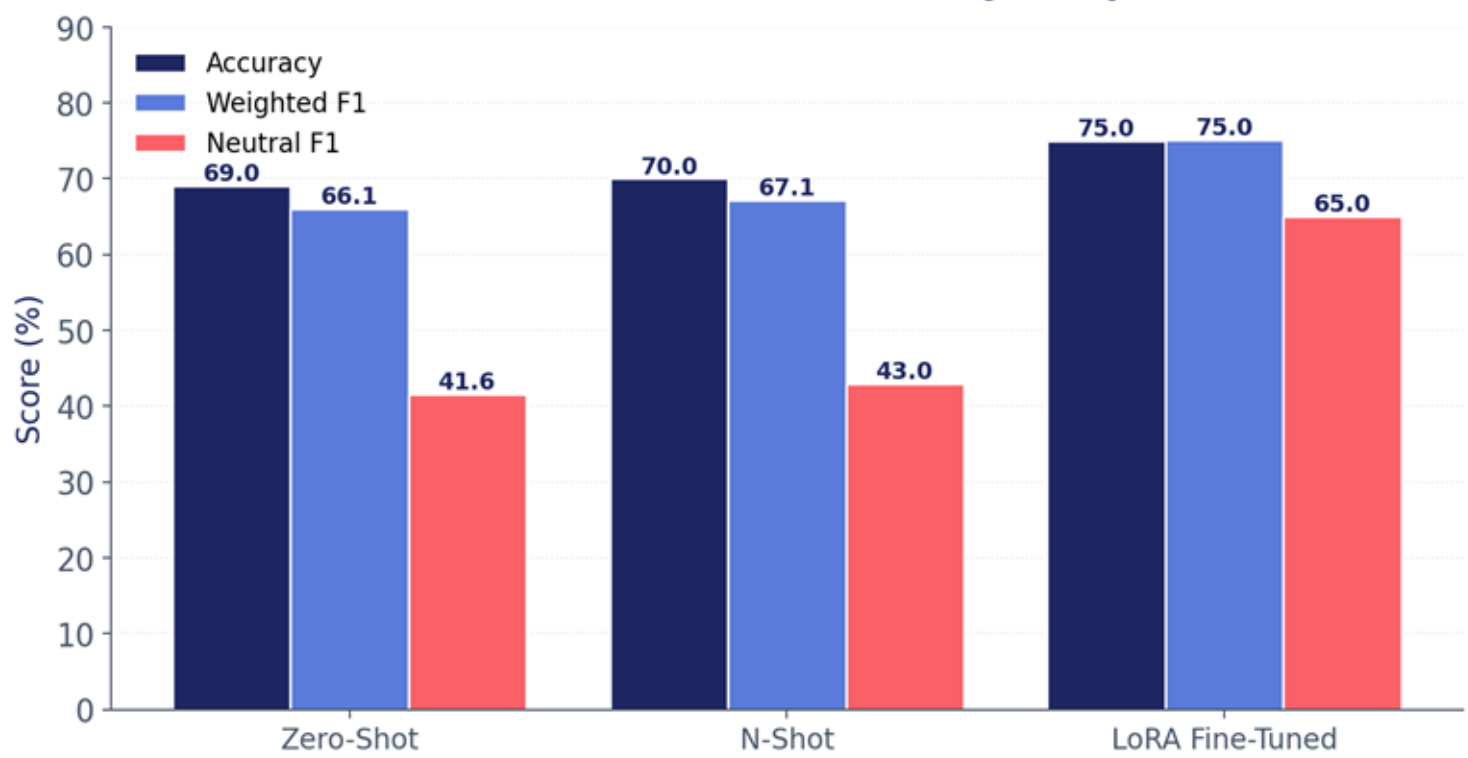
LoRA fine-tune

1500 balanced examples · $r=8$

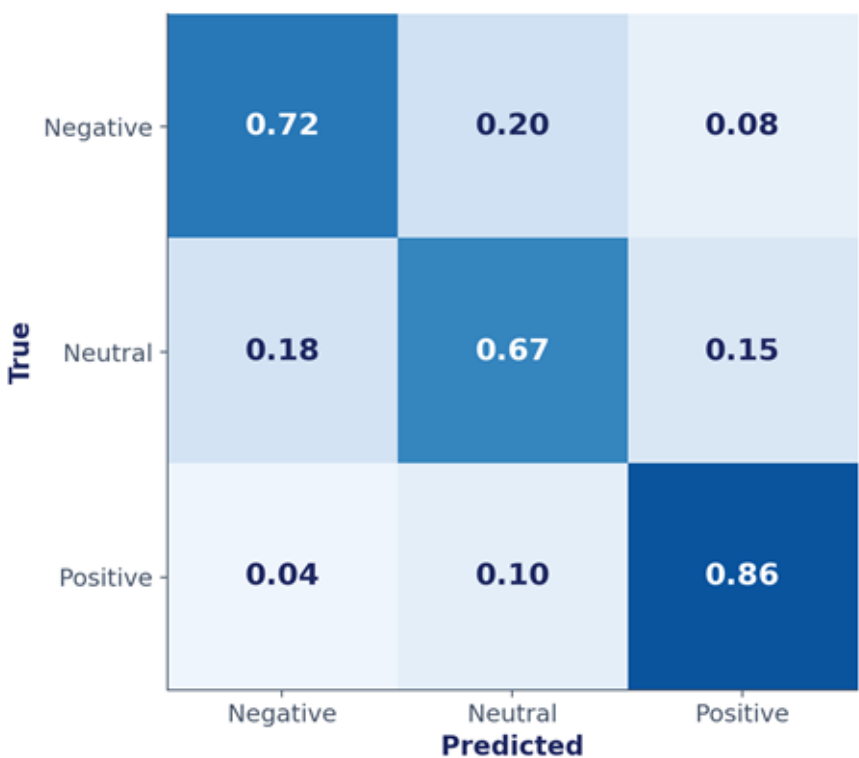
Model 1 — Evaluation

LoRA wins, and the gain is statistically real

Sentiment classifier — three-way comparison



LoRA confusion matrix (normalized by row)



+6 pts

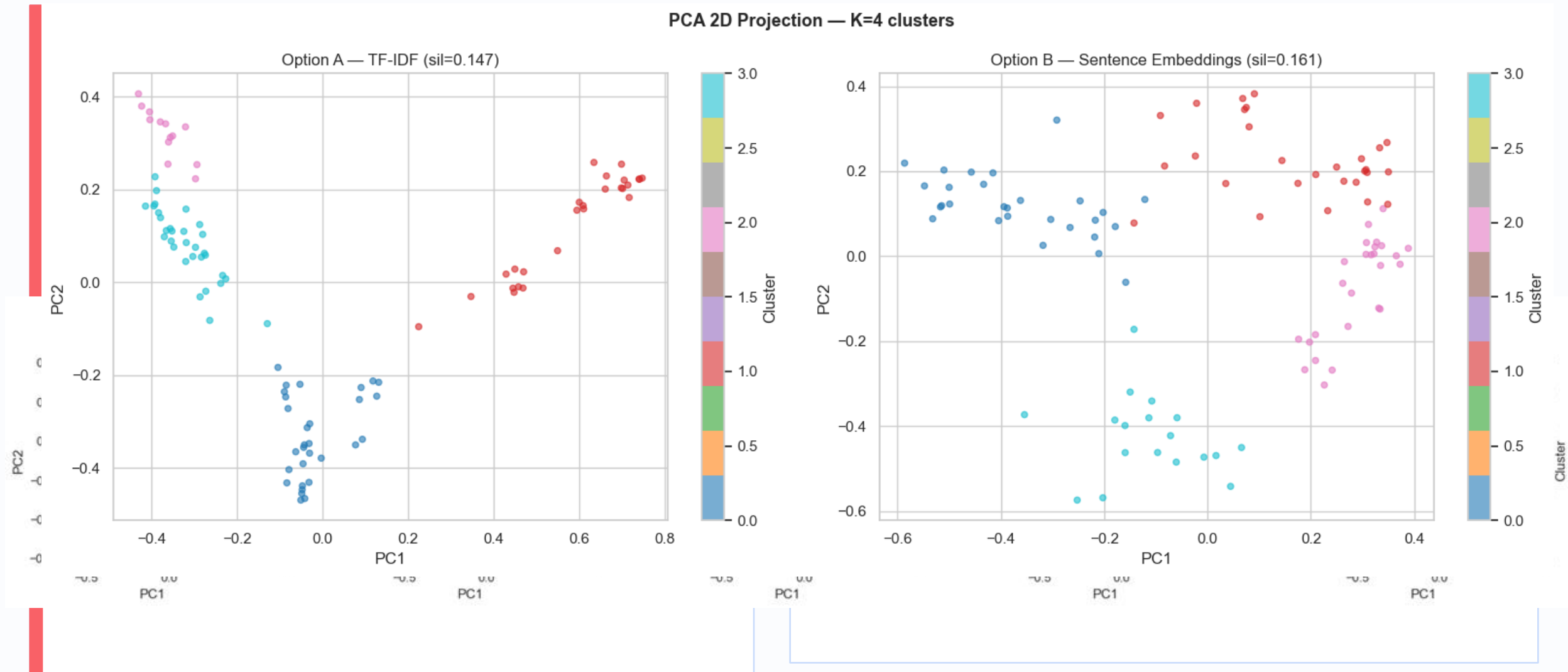
accuracy gain (LoRA)

+0.23 F1

neutral class recovery

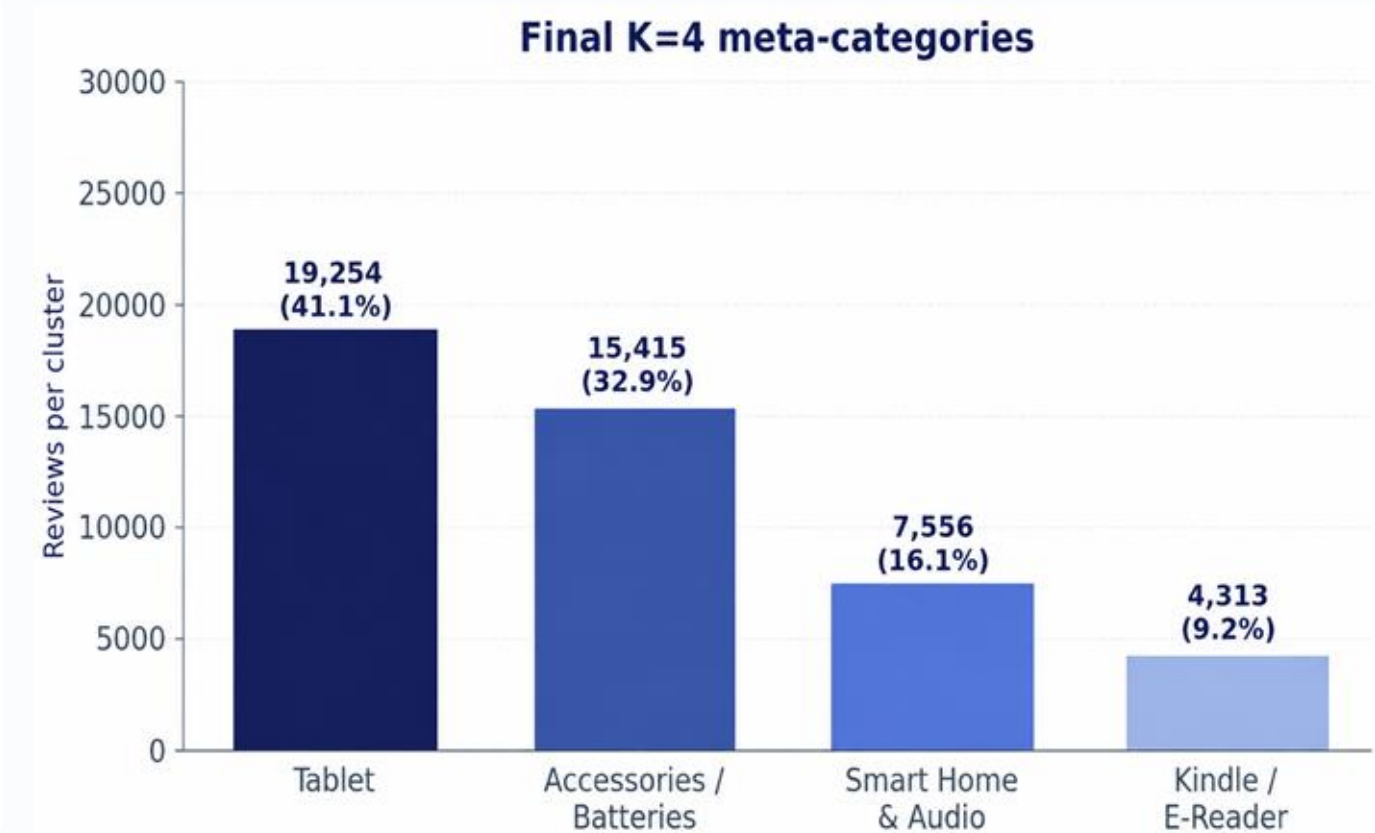
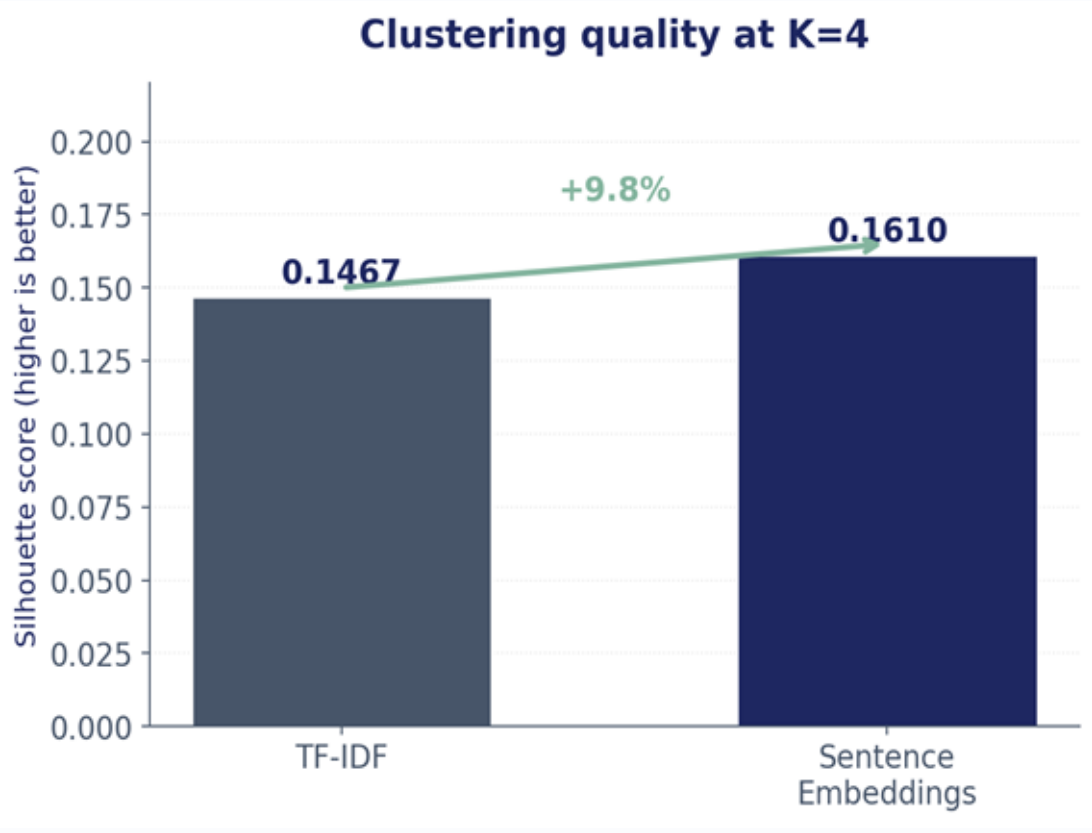
Model 2 — Category Clustering

Compress noisy product taxonomy into 4–6 meta-categories



Model 2 — Evaluation

Sentence embeddings won. Then we picked K=4 for humans.



Accessories / Batteries

19,254 reviews

Kindle / E-Reader

15,415 reviews

Tablet

7,556 reviews

Smart Home & Audio

4,313 reviews

Model 3 — Article Generation

From cluster profile to a publishable recommendation article

Self-distillation: model teaches itself

Why? No human-written reference articles exist for this dataset



Risk: model imitates its own bias — high ROUGE $\neq$ high quality

Base model

Qwen 2.5-1.5B-Instruct

Adapter

LoRA $r=16 \cdot q/k/v/o_proj$

Training data

Self-distilled references
(4 categories)

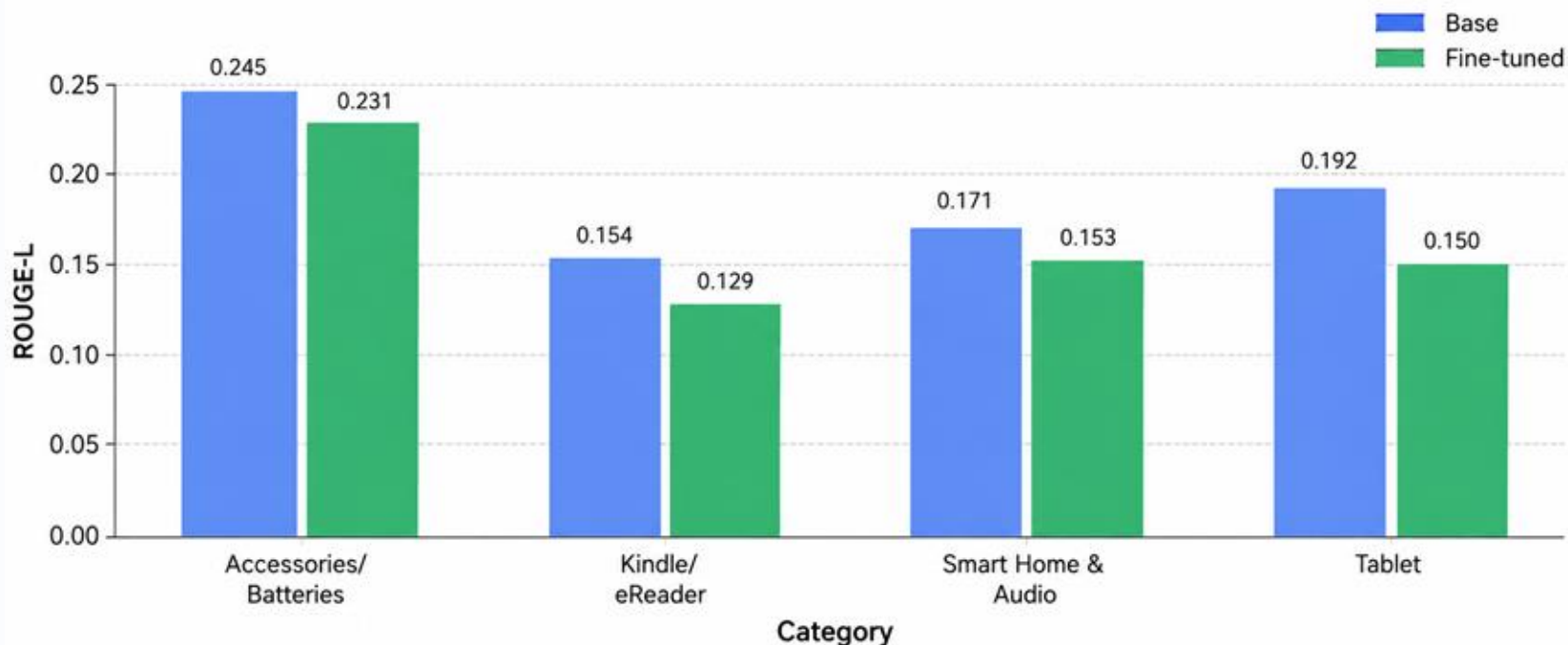
Eval metric

ROUGE-1 / 2 / L
vs held-out reviews

Model 3 — Evaluation

Mixed picture: gains on 2 of 4 categories

ROUGE-L: Base vs Fine-tuned Generator



Honest read

- Average ROUGE-L decreased from 0.191 to 0.166 (-0.025)
- All 4 categories scored lower after generator LoRA in this rerun
- Grounding checks: 0 hallucination flags; average product grounding 75%
- Interpretation: fine-tuning changed wording/structure but did not improve lexical overlap

Average ROUGE across all categories →

Base 0.191

→ Fine-tuned 0.166 (-0.025)

Takeaway

What we built, what worked, what didn't — and the medal

75.0%

Sentiment Accuracy

*LoRA fine-tuned
(+6 pts vs zero-shot, $p=0.036$)*

K=4

Clean Meta-Categories

*Sentence embeddings beat TF-IDF
silhouette 0.161*

-0.025

ROUGE-L decrease on best cat.

What worked

- LoRA → +6 pts accuracy, statistically significant
- Strict no-leakage train/test split + handoff checks
- K=4 categories were usable
- Gradio demo connects the pipeline

What's still open

- No classical baseline (LogReg + TF-IDF) for Task 1
- ROUGE alone doesn't capture article quality (Manual article review still matters)
- Self-distillation can amplify base-model bias